Vision-Based Flow-Front Assessment in Vacuum-Assisted Resin Transfer Molding

J.C. Vaught[1], Marshall Pigford[2], Declan Johnson[3] and Darun Barazanchy[4]
University of South Carolina, Columbia, SC, 29208, USA

Vacuum-Assisted Resin Transfer Molding (VARTM) operators read the visible advance of the resin flow front under the transparent vacuum bag as their primary process indicator. Prior vision systems for VARTM and Liquid Composite Molding use the same classical visual primitives, namely reference-frame differencing, Otsu thresholding, and morphological cleanup, but each system uses a different subset, no system reports the joint and individual contribution of those primitives, and no labeled video benchmark exists against which to attribute that contribution. This work presents an integrated classical computer-vision pipeline that combines a peak-brightness reference, a darken-only difference, region-of-interest restriction, dynamic-lag reference selection, morphological cleanup, and persistence-based temporal locking, evaluated on a fifty-five-frame hand-labeled subset spanning eleven distinct VARTM infusion runs. The integrated configuration reaches mask Intersection-over-Union of $X.XXX$ (95 % bootstrap confidence interval $[a, b]$) and boundary F_1 of $Y.YYY$ (CI $[c, d]$). A per-sample component-removal ablation on the same subset characterizes the marginal IoU effect of each named primitive, both as an eleven-sample mean and broken down per sample so that primitives whose contribution depends on infusion regime are visible rather than averaged away. The pipeline runs at thirty frames per second on a single CPU and at K frames per second on a CUDA implementation across the eleven samples.

I. Nomenclature

F_t	=	converted-colorspace frame at time index t
G_t	=	reference image at time index t
$G_t^\star$	=	effective reference (peak map when peak-reference enabled, otherwise channel-zero of G_t)
P_t	=	per-pixel peak-brightness map at time t
D_t	=	per-pixel response field at time t
R	=	binary region-of-interest mask
H, W	=	frame height and width in pixels
C	=	channel count of the working colorspace
α	=	exponential-moving-average factor for running-reference mode
κ	=	sqrt-area growth factor estimated from calibration frames
ρ	=	target wet-fraction for dynamic reference mode
λ	=	user lag scale factor for dynamic reference
τ_t	=	elapsed time at frame index t in seconds
$\Delta\tau_t$	=	dynamic lag in seconds for the reference frame
w_c	=	channel weight in the delta computation

[1]Graduate Research Assistant, Department of Mechanical Engineering, AIAA Student Member.
[2]Undergraduate Research Assistant, Department of Computer Science and Engineering, AIAA Student Member.
[3]Graduate Research Assistant, Department of Mechanical Engineering, AIAA Student Member.
[4]Research Assistant Professor, Department of Mechanical Engineering, AIAA Member.

N_{calib}	=	number of calibration frames before dynamic reference activates
f	=	video frame rate in frames per second
IoU	=	Intersection-over-Union mask agreement score
F_1	=	boundary F_1 score, mean across pixel tolerances $\tau \in \{1, 3, 5\}$

II. Introduction

Vacuum-Assisted Resin Transfer Molding (VARTM) is the dominant process for one-off composite laminates from research panels through small-series aerospace structural parts, and the principal observable signal available to the operator during a run is the visible advance of the resin flow front under the transparent vacuum bag [1].

The position, shape, and rate of that front carry process-quality information that other, more instrumented modalities recover only indirectly. Distributed dielectric arrays [2] and area-sensor grids [3] produce excellent local wetness measurements but require sensors embedded in the lay-up, and fiber-optic frequency-domain reflectometry [4] demands instrumentation no operator on a research bench will install for a one-off panel. Direct video of the bag is the one modality that costs nothing extra, sees the entire laminate at once, and is already recorded in most labs as a procedural artifact.

Prior camera-based VARTM and Liquid Composite Molding (LCM) systems agree on the basic recipe. Difference the live frame against a reference image of the dry preform, threshold the response into a binary candidate mask, and clean the mask with morphology [5–7]. They disagree on every detail that controls how well the recipe works. Some systems clip the difference so that only darkening counts as wetting and others do not [7]. Some track a per-pixel running maximum as the reference and others pin a single early frame [5]. Some impose a temporal-locking rule that holds a positive detection through transient dropouts and others operate frame-by-frame [8]. None of these systems reports the joint contribution of the components it uses, the marginal cost of each component it discards, or the agreement of the resulting mask against a labeled video benchmark, because no shared labeled benchmark exists. The practitioner is left to take the recipe at face value.

The work is hard for three concrete reasons that have to be addressed by anything claiming to do it. The vacuum bag produces strong specular reflections from lab lighting that look bright in every channel and shift on every operator move. The bag wrinkles and creases that pre-date the front are static dark features that do not represent wet fabric and must not be mistaken for it. The lighting itself drifts across a multi-minute infusion as lamps warm up and the operator moves equipment. A pipeline that fails to address any one of these will mark unwetted fabric as wet on most frames of most infusions. Prior camera-based systems individually defeat one or two of the three. None reports which component defeats which failure mode, and none reports the price paid for leaving any of them out.

This paper presents an integrated classical-computer-vision pipeline that combines, in one configuration, the visual primitives the prior systems use in isolation. The pipeline restricts the comparison to a region of interest, projects the working frame into the lightness channel of CIELAB, accumulates a per-pixel peak-brightness reference that absorbs lighting drift, computes a darken-only difference that rejects specular highlights, dynamically lags the reference frame against fill rate, thresholds the response with Otsu, cleans the mask with morphological closing and opening followed by connected-components filtering, and locks pixels whose wet label has persisted for several consecutive frames. On a 55-frame hand-labeled subset spanning eleven distinct VARTM infusion runs, the integrated configuration reaches mask Intersection-over-Union (IoU) of $X.XXX$ with a 95 % bootstrap confidence interval of $[a, b]$ and boundary F_1 of $Y.YYY$ with confidence interval $[c, d]$. A per-sample component-removal ablation on the same subset reports the marginal IoU effect of disabling each named primitive in turn. The reported effects are not uniform in sign or magnitude across the eleven samples, which is the empirical content of the ablation rather than a defect, since a primitive that helps a long operator-view recording can be neutral or counterproductive on a high-frame-rate clip whose dynamics it was not tuned for. The pipeline runs at 30 frames per second on a single central-processing-unit (CPU) thread, and a CUDA implementation reaches K frames per second on the same eleven samples.

The contributions of this paper are the following.

1. An integrated VARTM flow-front segmentation pipeline that combines peak-brightness reference, darken-only difference, ROI restriction, dynamic-lag reference selection, Otsu-plus-offset thresholding, morphological

cleanup, and persistence-based temporal locking in one configuration, with mean IoU $X.XXX$ and boundary F_1 $Y.YYY$ reported with 95 % bootstrap confidence intervals on a 55-frame hand-labeled subset spanning eleven distinct infusion runs.

2. A component-removal ablation that quantifies the marginal IoU effect of each named primitive on the same 55-frame subset, both as an eleven-sample mean and as a per-sample breakdown. The breakdown is reported in full because the effect of any one primitive is regime-dependent, so the joint composition is justified by the per-sample evidence rather than by a single mean.

Section 2 places this work among the three families of prior work that touch the problem. Section 3 walks each component of the pipeline and points forward at the ablation evidence. Section 4 documents the labeling protocol and the eleven-sample dataset. Section 5 reports the integrated configuration's agreement against the labeled subset and the per-component ablation. Section 6 reports CPU and CUDA throughput. Section 7 discusses failure modes and design tradeoffs. Section 8 concludes.

III. Related Work

Prior work on Vacuum-Assisted Resin Transfer Molding (VARTM) and the broader Liquid Composite Molding (LCM) family touches the flow front from three different directions. A first body of work uses flow-front information to act on the process through closed-loop control or fault response. A second body observes the process to characterize it, typically with embedded sensors or auxiliary cameras for offline study. A third body models the underlying Darcy physics or inverts it from sparse measurements. The work in this paper sits at the seam of all three because the integrated pipeline produces the per-frame full-field flow-front mask that the first family assumes is available, that the second family currently approximates with sensor grids, and that the third family typically receives only as a handful of digitized contours from a manual annotation pass. The remainder of this section addresses each family in turn and closes with the specific gap that the integrated pipeline fills.

A. Use: vision and sensors that act on the VARTM process

Closed-loop VARTM controllers have a long lineage. Multi-segment injection-line work coupled point sensors with a finite-element flow model so that valves could be re-sequenced in real time to chase the flow front toward starved regions and away from race-tracks [9]. The control loop is genuinely real time and the empirical results are convincing for plate geometries, but the feedback signal is a sparse vector of resin-arrival times at instrumented locations, which means dry-spot detection only works when a sensor happens to lie inside the dry region. The integrated pipeline differs by treating every pixel inside the bag as a potential sensor, so dry-spot evidence is not gated on instrumentation density.

A more recent line of work replaces point sensors with a top-mounted camera. Lekanidis and Vosniakos built a webcam-plus-microcontroller rig for VARI that segments the resin flow front with a classical Matlab pipeline (Gaussian blur, grayscale conversion, contrast stretch, Otsu binarization, closing on a 10-pixel disk, Sobel edge detection, opening with a 120-pixel area threshold, dilation) and toggles inlet valves on the resulting front position [5]. Almazán-Lázaro and co-workers close a similar loop on resin infusion, regulating flow-front velocity to an experimentally-determined optimum with a PID over a pipeline that combines region-of-interest cropping, Scaramuzza-style lens-distortion correction, histogram equalization, first-frame absolute differencing, grayscale conversion, a 5-by-5 mean filter, Sobel-gradient segmentation, paired erosion and dilation, and small-area removal [6,7]. Both systems demonstrate that vision alone can drive VARTM control, but neither reports the joint or individual contribution of the segmentation primitives they use, neither evaluates the segmentation against a labeled video benchmark, and both treat the segmentation step as a black box that succeeds or fails as a unit. The work in this paper differs by combining the same family of primitives in one configuration and reporting the agreement of the joint configuration and the IoU cost of disabling each component on a labeled subset spanning eleven distinct infusion runs.

Adjacent to control are vision-based defect-detection systems for composites more broadly [10]. These pipelines now lean almost entirely on convolutional or transformer detectors trained on hand-curated defect crops, and a recurring bottleneck across that survey is the cost of pixel-accurate labels. The work in this paper is upstream of that bottleneck. The integrated classical pipeline produces per-pixel front masks deterministically and characterizes which components are responsible for which IoU points, which gives a downstream learned detector [11,12] a labeled-data starting point whose error structure is decomposable rather than monolithic.

B. Observe: sensor-based and vision-based process characterization

A second family treats the flow front as something to be measured rather than acted on. Konstantopoulos and co-authors review the in-line sensing landscape and group the field around dielectric, thermal, ultrasonic, fiber-optic, and electrical-resistance sensors [1]. Dielectric tracking by Yenilmez and Sozer and others gives unsaturated and saturated front positions at each electrode pair, with an installation cost that is hard to beat, but the spatial resolution of the resulting front is limited by the electrode pitch [2]. Matsuzaki and co-workers dramatically improve coverage with an area-sensor array that tiles a polyimide film into a dense capacitive matrix and recovers the impregnated-area ratio per cell, which is the closest sensor-side analogue to a full-field flow-front mask [3]. The resulting front is, however, still a quantized grid that is destructive to instrument inside production tooling and that adds permeability to the stack-up.

Fiber-optic strategies use Fiber Bragg Grating arrays or Optical Frequency Domain Reflectometry to time-stamp resin arrival at every grating along an embedded fiber, and recent distributed-sensing implementations push toward life-cycle structural health monitoring of large CFRP parts [4]. These methods deliver excellent temporal precision and can survive infusion and cure, but the same coupling between sensor density and front fidelity reappears, since arbitrary front shapes between gratings are interpolated, not measured. A small but growing thread fuses dielectric and visual data with learning-based fusion. The recent AI-based dielectric-plus-visual monitor by Esposito and co-workers detects and tracks the front with a YOLO detector trained on labeled video and uses dielectric channels as a redundant signal [8]. That pipeline is closest in spirit to the front producer described here, but it is reported on a single-fabric study, it does not report which preprocessing components carry which IoU points, and the detector is trained rather than classical. The work in this paper differs by combining the classical primitives the prior systems use in isolation, evaluating the joint configuration on a labeled subset spanning eleven infusion runs, and reporting the marginal IoU effect of each component.

Stieber and co-workers show that learning a flow-front “image” from a sparse pressure-sensor grid is a tractable surrogate task and report time-step accuracy near 92 percent on simulated injection runs [13]. A follow-on extends the same approach to material-property inference under a simulation-to-real transfer pretext and shows IoU around 0.50 with transformer backbones [14]. These are excellent demonstrations of what a network can learn given ground-truth flow-front maps. They presume the maps exist, and they obtain them from FEM injection simulations rather than from real video. The work in this paper addresses the upstream half of that pipeline by producing per-pixel front masks from real VARTM video that such networks would need for fine-tuning, and by reporting which components of the upstream segmenter contribute which IoU points.

C. Model: physics-based forward and inverse flow-front formulations

The third family models the front analytically or numerically. Forward Darcy-flow simulators for VARTM now routinely couple a depth-averaged Darcy solver to a level-set front with high-permeability-medium layers and progressive-saturation effects [15], and end-to-end VARTM simulation with reinforcement compaction and post-filling pressure relaxation has been validated experimentally for plate and stiffened-plate geometries [16]. These models predict the flow front given a permeability field and a boundary condition. The inverse direction is harder. Comas-Cardona and co-authors recover an in-plane permeability field by coupling optical areal-weight measurement with central-injection front observation, and similar work by Di Fratta, Klunker, and Ermanni inverts pressure-sensor traces against an analytic front model to update injection parameters online [17,18]. These methods are mathematically careful, but they all depend on having the front geometry as input. In Comas-Cardona that geometry comes from a manually thresholded plate experiment, and in the sensor-based works it comes from arrival-time interpolation. The integrated pipeline lowers the cost of populating that input by producing per-frame, per-pixel front masks directly from the production video stream that is already being recorded for documentation.

Recent work has begun to merge model and learning. Park and co-authors couple electromechanical PZT signals to a tree-plus-GAN model that both identifies the live front and forecasts its near-future position [19], and physics-informed neural permeability inversion has been explored in synthetic Darcy benchmarks [20]. Both are promising for closed-loop VARTM, and both inherit the same dependency on a labeled flow-front signal during training. Without an open, retargetable source of that signal in real video, every group reproducing or extending these methods has to instrument a new bench.

D. Building blocks: classical CV foundations

The integrated pipeline stays deliberately within well-understood classical operators. Otsu's bimodal threshold provides the wet-versus-dry separation in each frame [21], the Suzuki-Abe topological-border-following algorithm provides hierarchically consistent contour extraction over the resulting binary [22], and the Kim et al. codebook background model handles the slow illumination drift typical of multi-hour infusions without retraining [23]. None of these primitives is novel. The contribution of this paper is the joint composition, the per-stage parameterization, and the per-component evidence that the joint matters on a labeled multi-mold subset. The recent material-defect-detection survey notes that the absence of characterized preprocessing pipelines is a primary bottleneck for deploying deep models in composite quality control [10], and modern detectors such as YOLO illustrate why that bottleneck matters because they are starved for in-domain pixel labels rather than for capacity [24].

E. Where this work fits

The three families above all touch the flow front, but none of them reports the question this paper answers. The control family treats segmentation as a black box and reports downstream control quality, not segmentation quality. The observation family reports excellent agreement at sensor electrodes and gratings, but interpolates between them rather than measuring per-pixel front geometry from video. The modeling family assumes a per-pixel front exists and consumes it, which is precisely the artifact the upstream segmenter must produce. Within the camera-based subset of the observation family, prior systems differ in which subset of the available primitives they use and report only end-to-end behavior on a single mold. None reports the joint contribution of the components in use, none reports the marginal cost of any component left out, and none evaluates against a labeled video benchmark spanning more than one mold. This paper supplies that evidence on eleven distinct infusion runs and on a 55-frame hand-labeled subset.

IV. Method

A. Pipeline overview

The integrated pipeline is built from primitives that each prior camera-based VARTM and LCM system uses in some subset. None of the primitives is novel in isolation. The empirical question this paper supplies evidence for is what each primitive contributes to the joint result on a labeled multi-mold subset, and the method described below is the configuration that supports that question.

The detection algorithm treats each video frame as a comparison against a reference image of the dry preform. Where resin has wetted the fabric, the local appearance darkens, and the absolute change is largest near the advancing flow front. The pipeline first checks for camera motion against the previous frame and, when motion crosses a configured threshold, registers the live frame back into the reference coordinate system before any difference is computed. The registered frame is optionally smoothed with a pre-delta blur whose output feeds both the running peak map and the difference computation, so the reference and the comparison see the same bandwidth-limited input. The pipeline computes the per-pixel difference inside a rectangular region of interest, clips it to admit only darkening so that specular highlights from the vacuum bag are rejected, normalizes the resulting field, thresholds it into a binary candidate mask, cleans the mask with morphological closing and opening, removes small connected components, and finally locks pixels whose wet label has persisted for several consecutive frames. The locked mask is the canonical output, the optional heatmap renders the underlying delta field for visual diagnosis, and the optional overlay draws the mask boundary onto the original Vacuum-Assisted Resin Transfer Molding (VARTM) frame for human review. Figure Fig. 1 shows the sixteen stages in order.

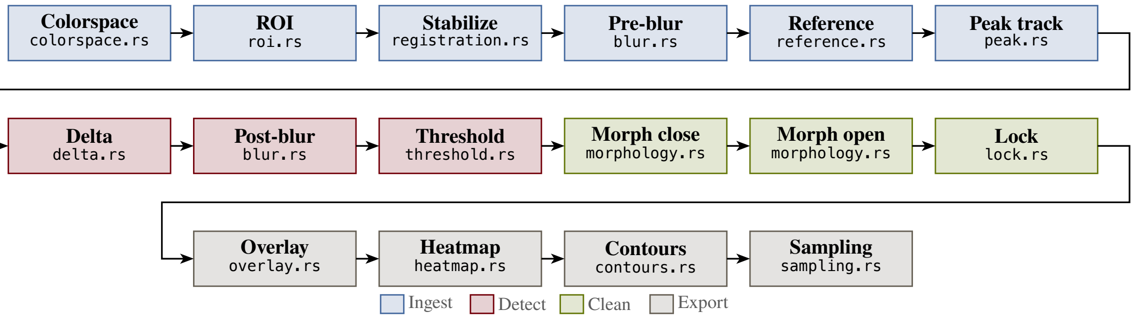

Fig. 1 Sixteen-stage integrated pipeline. Each frame flows from decode through an optional camera-shift registration and pre-delta blur, then through an appearance-difference comparison against a chosen reference, a post-delta blur, a threshold, paired morphological closing and opening passes, and finally a temporal lock that stabilizes the boundary across frames. The same mask drives the binary, heatmap, and overlay renderers.

The pipeline is deliberately classical because the empirical claim of this paper is per-component. Every stage is an explicit function of one frame, the chosen reference image, and a small set of scalar parameters, which is the property that makes the component-removal ablation in Section 5 well defined. A learned segmenter would conflate the contributions of the components inside it, and the question this paper asks would not be answerable.

B. Frame decode and colorspace conversion

The first stage decodes each Blue-Green-Red (BGR) frame from the input video. The second projects that frame into a working colorspace. The pipeline supports four options, namely the International Commission on Illumination (CIE) 1976 $L^*a^*b^*$ colorspace (CIELAB), Red-Green-Blue (RGB), Hue-Saturation-Value (HSV), and 8-bit grayscale. The integrated configuration uses CIELAB because for the resin-and-fabric combinations in the eleven samples evaluated here, wetting primarily darkens the lightness channel L^* rather than shifting chrominance, so the single-channel L^* projection used in the difference computation captures most of the signal. The choice is regime-dependent rather than universal. A pigmented resin or a colored fabric could shift the balance of evidence into chrominance and make RGB or HSV preferable, which is why the alternatives remain selectable. The eleven-sample mean and per-sample breakdown of the colorspace effect are reported in Table Table 12. We denote the converted frame at index t by $F_t \in \mathbb{R}^{H\times W\times C}$, where C is the channel count of the chosen colorspace.

Colorspace projection placeholder. Four-panel grid showing the same canonical input frame projected into each of the four supported colorspaces. Column 1: raw BGR. Column 2: CIELAB lightness L^* (used by the integrated configuration). Column 3: HSV value channel. Column 4: 8-bit grayscale. Each panel rendered with a perceptually uniform colormap so wet-vs-dry separability per projection is visible. A bottom strip shows the per-projection histogram for the laminate region of interest with the wet and dry modes annotated. Regenerated from the new ablation runs.

Fig. 2 The four working-colorspace projections supported by the pipeline.

Table 1 Working-colorspace options selectable via `--colorspace`.

Option	Description
CIELAB	CIE 1976 $L^*a^*b^*$. Wetting primarily darkens L^* for the resin-and-fabric combinations evaluated in this paper.
RGB	Red-Green-Blue. Useful when chrominance shifts under wetting are visible per channel, e.g. pigmented resin.
HSV	Hue-Saturation-Value. Saturation tracks wet-out separately from luminance, useful under variable lighting.
Grayscale	8-bit luminance projection. Cheapest but loses chrominance evidence entirely.

C. Region of interest

A rectangular region of interest excludes the part frame, manifold flange, and bag wrinkles outside the laminate. The user supplies four fractional margins, one per edge, each clamped to the closed interval from zero to forty-nine hundredths of the corresponding side, and the algorithm builds a binary mask R that is one inside the resulting rectangle and zero elsewhere. A single configuration parameter sets all four margins symmetrically, with per-edge overrides available. All subsequent stages operate only on pixels where $R = 1$.

ROI crop placeholder. Single canonical input frame with the rectangular ROI mask R drawn as a translucent garnet overlay over the laminate; the manifold flange and bag wrinkles outside the rectangle are visibly excluded. A small inset in the corner shows the four fractional margins (`roi-margin-top`, `-bottom`, `-left`, `-right`) used in the integrated configuration. The hatched region outside the rectangle is the area all subsequent stages skip.

Fig. 1 ROI mask R applied to a canonical input frame. Pixels outside the rectangle are excluded from every subsequent stage.

D. Camera-shift detection and registration

A bumped tripod, a thermal expansion of the rig, or a hand brushing the camera in mid-run shifts the projected position of every laminate pixel by some vector that has nothing to do with wetting. The reference frame stops aligning with the live frame, the difference field lights up along high-contrast laminate edges, and the threshold catches that as wet. The pipeline detects the shift and corrects it before the difference is computed. For each frame, a single phase-correlation step on a fixed-resolution downsample of the working channel of the previous and current frames returns a translation (d_x, d_y) and a confidence score. When either the per-frame magnitude $\sqrt{d_x^2 + d_y^2}$ or the cumulative drift across a five-frame rolling window exceeds the configured threshold, an iterative-coplanar-correlation refinement fits a translation or affine warp W_t on a static-edge mask of the current ROI, and the live frame is warped through W_t into the reference coordinate system before all downstream stages. The static-edge mask is recomputed at each shift event so the registration is driven by mold and frame edges that do not move with the wet front rather than by the wet region itself. The peak map is either discarded (`peak-on-shift = reset`, used by the integrated configuration), so the post-warp pixels do not accumulate against pre-warp brightness, or warped through W_t into the new coordinate system (`peak-on-shift = warp`), so the running maximum is preserved at the cost of carrying any residual registration error into the peak.

Camera-shift event placeholder. Three panels of the bumped-tripod event near frame 71 of the canonical reference video. Left: pre-event frame (frame 70). Center: post-event frame (frame 72) overlaid translucently on the pre-event frame with the phase-correlation translation vector drawn as an arrow. Right: the post-event frame after the iterative-coplanar- correlation refinement warps it through W_t back into reference coordinates, with the static-edge mask overlaid showing which mold-and-frame edges drive the warp. A sub-panel below plots the rolling five-frame cumulative motion magnitude across the run with a marker on frame 71. Regenerated from the new ablation runs.

Fig. 3 A bumped-tripod event in the canonical reference video. The phase-correlation step detects the shift, the iterative-coplanar-correlation refinement warps the live frame back into reference coordinates, and the rest of the recording shows only sub-pixel residual motion.

Table 2 Camera-shift handling options. The integrated configuration uses `motion-model = affine` and `peak-on-shift = reset`.

Parameter	Option	Description
`motion-model`	translation	Two-degree-of-freedom translation warp; cheaper to fit, recovers pure tripod nudges.
`motion-model`	affine	Six-degree-of-freedom affine warp; tolerates small rotations and shears in addition to translation.
`peak-on-shift`	reset	Discard the peak map at registration and restart accumulation from the warped frame.
`peak-on-shift`	warp	Warp the peak map itself through W_t into the new coordinate system; preserves the running maximum at the cost of carrying residual registration error.

E. Peak-brightness reference

The pipeline accumulates a running maximum of the working channel across all frames seen so far. When the pre-delta blur is enabled the maximum is updated from the blurred working channel rather than the raw channel so the peak map and the delta input share the same bandwidth, which prevents a one-pixel halo of speckle from shifting the peak above the eventual delta input and creating a phantom darkening signal. The motivation for the running maximum is that, on infusions where the dry preform is the brightest state the fabric reaches in the working channel, a per-pixel maximum tracks lighting drift caused by lamp warm-up and bag deformation rather than treating that drift as wetting evidence. Figure Fig. 4 illustrates the regime on one tracked pixel of the canonical input clip. The raw L^* value drifts upward as the lamps stabilize, the running peak P tracks that drift, and the front arrival separates as a sharp drop of more than thirty units below the peak. A fixed reference at frame zero would have included the post-warm-up lift in its difference. The assumption that wetting is the only event that lowers the peak does not hold universally. A specular flash that drives the peak above its long-term value, or a fabric whose wet state is brighter than its dry state for some channel, can produce a peak map that misrepresents subsequent wetting evidence. The IoU effect of disabling the primitive on the eleven-sample subset is reported in Table Table 12.

Peak-reference trace placeholder. One pixel of the canonical reference video tracked across all frames; raw L^* in garnet, running peak P in atlantic, with the front-arrival drop annotated. Regenerated from the new ablation runs.

Fig. 4 One pixel of the first canonical input clip tracked across all 706 frames. The raw CIELAB lightness L^* drifts up by roughly twenty units before the front arrives, and the running peak P absorbs that drift. When the resin reaches the pixel, L^* collapses by more than thirty units below the peak, which is the signal the difference stage detects. A single fixed reference at frame zero would have treated the warm-up brightening as a (negative) wetting event.

The peak map can be disabled through a single configuration parameter. The "no peak-brightness reference" row of Table Table 12 reports the IoU effect of doing so on each of the eleven samples, with the per-sample breakdown showing that the effect is not constant across infusion regimes.

F. Reference selection

The reference image G_t used for the difference computation has five modes. The first-frame mode pins $G_t = F_0$ for every t. The running mode updates an exponential moving average $G_t = (1-\alpha)G_{t-1} + \alpha F_t$ with $\alpha = 0.05$. The previous-frame mode uses a fixed-offset history $G_t = F_{t-k}$, with the buffer bootstrapped from F_0 until $t > k$. The absolute mode pins G_t to a user-specified absolute frame index. The dynamic mode adapts the lag online from a square-root-area growth model fit to the early frames. The integrated configuration uses the first-frame mode, which, combined with the peak map above, gives a piecewise-static reference whose only adaptation is the peak update.

The dynamic mode warrants a closer look because the integrated configuration does not use it for the headline numbers and disabling it is one of the rows in Table Table 12. For the first N_{calib} frames after detection bootstrapping, the pipeline records the wet area a_t inside the region of interest and the elapsed time $\tau_t = \frac{t-1}{f}$ in seconds, where f is the video frame rate. It then estimates a growth-rate factor κ as the median of $\frac{a_t}{\tau_t^{\frac{3}{2}}}$ across the calibration frames. The three-halves exponent is motivated by the area-growth law for radial Darcy infusion, where wetted area scales approximately as time to the three-halves power once flow is well established; the assumption holds approximately for the early-fill regime of the eleven samples in the labeling subset and breaks for race-tracking-dominated runs, which is why the dynamic-calibration window is sensitive to anomalous early frames as discussed in Section 7. Given κ and a target wet fraction ρ of the region-of-interest area $|R|$ (held at 0.2 in the integrated configuration), the dynamic lag $\Delta\tau_t$ in seconds that places the reference at the moment when the wet area was a fraction ρ of the current area is

$$\Delta\tau_t = \lambda \cdot \left[\left(\frac{\rho\,|R|}{\kappa} + \sqrt{\tau_t} \right)^2 - \tau_t \right], \tag{1}$$

clipped to be non-negative and scaled by a user lag factor λ. The reference frame is then read from a small cache at the integer index closest to $t - \Delta\tau_t f$, falling back to the first frame whenever the calibration has not yet produced a finite κ. A linear-mode override replaces the sqrt-area growth fit with a linear lag schedule parameterised by `dynamic-lag-linear-start` and `dynamic-lag-linear-max`, which steps the reference frame back at a constant rate independent of the calibration estimate. The override is intended for diagnostic comparisons where the sqrt-area assumption is suspect, and a per-frame log of the chosen lag is written to `dynamic-lag-log` for post-hoc inspection.

Reference-mode comparison placeholder. Six-panel grid showing the same canonical input frame at fill position 50% under the five reference modes plus the dynamic linear-lag override. Top row: first-frame reference, running EMA reference, previous-frame fixed-offset reference. Bottom row: absolute-frame reference, dynamic sqrt-area reference, dynamic linear-lag override. Each panel shows the resulting D_t scalar field in the Turbo colormap so the reader can see how each reference mode selects different evidence on the same frame. The integrated- configuration mode (first-frame combined with the peak map) is bordered in garnet. Regenerated from the new ablation runs.

Fig. 5 The five reference-selection modes the pipeline supports plus the linear-lag dynamic override. The integrated configuration uses the first-frame mode combined with the peak map.

Table 3 Reference-selection options selectable via `--ref-mode`.

Option	Description
first	Pin $G_t = F_0$ for every t. Combined with the peak map, used by the integrated configuration.
running	Exponential moving average $G_t = (1-\alpha)G_{t-1} + \alpha F_t$.
previous	Fixed-offset history $G_t = F_{t-k}$.
absolute	Pin G_t to a user-specified absolute frame index.
dynamic (sqrt-area)	Adapt the lag online from a square-root-area Darcy growth fit on N_{calib} calibration frames per Eq. Eq. (1).
dynamic (linear)	Constant-rate lag schedule, independent of the growth fit; intended for diagnostic comparisons.

G. Delta computation

Stage seven produces the per-pixel scalar field D_t that drives all downstream decisions. The integrated configuration's darken-only mode operates on the working (first) channel and records how much darker the frame is than its reference,

$$D_t(y, x) = R(y, x) \cdot \max(0, w_0 \cdot (G_t^\star(y, x) - F_t(y, x, 0))), \tag{2}$$

where $G_t^\star$ equals the peak map P_t when the peak-reference mode is enabled and otherwise equals the channel-zero slice of G_t. The clip to non-negative values discards every pixel that becomes brighter than the reference, which removes specular flashes from the silicone vacuum bag and from condensation, neither of which are wetting events. A full-color mode replaces the difference with the channel-weighted Euclidean distance across all C channels and is intended for HSV and RGB workflows where chrominance shifts are diagnostic. The per-channel weights w_0, w_1, w_2 are exposed via `channel-weights` and equal $1, 1, 1$ in the integrated configuration; non-uniform weights are appropriate when one channel of the working colorspace carries the wetting signal more strongly than the others. Figure Fig. 6 shows the effect of the clip on a single frame; Table Table 12 reports the IoU cost of removing it across the eleven-sample subset.

Darken-only delta comparison placeholder. Two delta panels for the same frame: a naive Euclidean delta that lights up on the vacuum-bag specular highlight, and a darken-only delta that discards it. Regenerated from the new ablation runs.

Fig. 6 Same input frame, two delta computations. The naive Euclidean delta (centre) lights up bright on the vacuum-bag specular highlight to the upper right of the laminate. The darken-only delta (right) discards the highlight and isolates the front. The early-fill front in the lower part of the panel is visible in both, though more cleanly in darken-only.

H. Post-delta blur

The delta field is smoothed by the post-delta blur stage, which is a single function exposed by the `blur.rs` module and shared with the optional pre-delta blur of stage four. The user selects a kernel kind from {flat, gaussian, triangle, median, bilateral, none} together with a kernel size, written as a single specification of the form KIND[:SIZE]. The integrated configuration uses `gaussian:9`, a separable Gaussian of size $k_b = 9$ pixels (forced odd at runtime). Gaussian is appropriate when speckle is approximately white noise around the underlying response field; flat and triangle are exposed because some camera-and-bag combinations produce structured noise that the corresponding box or tent filter handles with less bias. Routing both the pre- and post-delta blurs through the same module keeps their behavior identical at matched specifications and ensures that the bandwidth-limited input the peak map sees in stage four is the same kind of bandwidth-limited input the threshold sees in stage nine.

Pre- versus post-delta blur placeholder. Two-row panel. Top row: working frame with no pre-delta blur (left), with $k_p = 5$ pre-delta blur (center), with $k_p = 9$ pre-delta blur (right). Bottom row: resulting D_t after each, all then post-delta- blurred with $k_b = 9$. The figure shows that pre-delta blur reduces speckle that would otherwise enter the peak map, while post-delta blur smooths the response field before thresholding. The integrated configuration disables pre-delta ($k_p = 0$) and applies post-delta with $k_b = 9$; both panels for that configuration are bordered in garnet.

Fig. 7 Effect of pre-delta blur (top) and post-delta blur (bottom) on the response field D_t.

Table 4 Blur kernels exposed by `blur.rs` and selected via the `KIND[:SIZE]` specification of `--pre-delta-blur` and `--blur`.

Option	Description
gaussian	Separable Gaussian kernel; appropriate when speckle is approximately white noise around the underlying response.
flat	Box (uniform-mean) filter; cheapest, biases edges more than gaussian.
triangle	Tent kernel; intermediate between flat and gaussian on edge handling.
median	Median filter; appropriate for salt-and-pepper noise.
bilateral	Edge-preserving cross-bilateral filter; preserves boundaries between wet and dry while smoothing inside each region.
none	No blur applied. Equivalent to specification `none` or kernel size zero.

I. Normalization and threshold

The smoothed field is rescaled to the byte range using a min-max linear rescaling, producing a $\tilde{D}_t$ that fits in eight bits and feeds both the threshold-selection stage and the heatmap renderer. Six thresholding modes are exposed through a single specification of the form KIND[:ARGS]. The four global modes share a single offset δ_τ that is added before binarization. Otsu's between-class variance method [21] recovers an automatic threshold τ_{otsu} over the full $\tilde{D}_t$ histogram and adds the offset, which is the integrated configuration's choice and works on infusions whose histograms are roughly bimodal. The Triangle method recovers τ_{tri} by the geometric construction of Zack and co-workers on the histogram [25], which is appropriate when one class dominates the histogram (typically early fill, where most pixels are dry). Manual mode uses a user-supplied absolute byte value τ_{man} plus δ_τ; percentile mode sorts the ROI pixels and recovers the p-th percentile by linear interpolation. The two adaptive modes, adaptive-mean and adaptive-gaussian, compute a per-pixel threshold from a $b \times b$ neighborhood mean (or Gaussian-weighted mean) minus a constant C, and bypass δ_τ because C already serves the same role. Adaptive thresholding is appropriate when the delta retains a low-frequency intensity gradient that the reference stage did not fully cancel, for instance under uneven side-lighting or vignette artifacts. The integrated configuration uses Otsu with $\delta_\tau = -30$, which biases the global threshold toward the wet class on infusions where the partially-wetted halo around the front sits below the bimodal split that Otsu finds. The bias is appropriate for the early-fill regime of the eleven labeled samples and not appropriate for every infusion. The Triangle, manual, percentile, and adaptive variants remain selectable for infusions whose histograms or intensity gradients fit those modes better; the integrated configuration does not exercise them, and the headline numbers of Section 5 are reported under Otsu plus offset.

Threshold-mode comparison placeholder. Six-panel grid of binary masks produced by each threshold mode on the same normalized response field $\tilde{D}_t$. Panels: Otsu plus offset (used by the integrated configuration), Triangle, manual at $\tau_{\text{man}} = 64$, percentile at $p = 70$, adaptive-mean with $b = 21$ and $C = 10$, adaptive-gaussian with $b = 21$ and $C = 10$. Above the grid, the histogram of the response is plotted with the Otsu and Triangle thresholds annotated. The integrated mode is bordered in garnet.

Fig. 8 The six threshold modes the pipeline supports applied to the same normalized response field.

Table 5 Threshold modes selectable via the `KIND[:ARGS]` specification of `--threshold`.

Option	Description
otsu	Between-class variance threshold over the histogram of $\tilde{D}_t$ [21], plus offset δ_τ.
triangle	Geometric construction of Zack and co-workers on the histogram [25], plus offset; appropriate when one class dominates the histogram.
manual	User-supplied absolute byte value τ_{man}, plus offset.
percentile	p-th percentile of ROI pixels of $\tilde{D}_t$ by linear interpolation, plus offset.
adaptive-mean	Per-pixel threshold from a $b \times b$ neighborhood mean minus constant C. Bypasses δ_τ.
adaptive-gaussian	Per-pixel threshold from a $b \times b$ Gaussian-weighted neighborhood mean minus constant C. Bypasses δ_τ.

J. Morphological close

Stage ten passes the binary mask through morphological closing with an elliptical structuring element of size k_m (default thirteen pixels). Closing welds neighbouring wet patches into a single front and fills small gaps where transient bag wrinkles muted the local response. The kernel size is the parameter that sets the spatial scale at which gaps are considered noise rather than real disconnections in the front; on the resolutions in the eleven labeled samples a 13×13 ellipse is large enough to bridge bag-wrinkle gaps and small enough to leave genuinely separate wet regions separate.

K. Morphological open and area filter

Stage eleven passes the closed mask through morphological opening with the same kernel and shape, removing specks below the kernel size that survived the closing pass. A connected-components labelling pass then discards any region whose pixel area is below a_{min} pixels (default 400), which removes the small islands that the morphology kernels are too small to suppress. Figure Fig. 9 shows the same frame at every step of stages nine through eleven.

Mask-cleanup montage placeholder. Five panels for one frame: the normalized response, the threshold output, after closing, after opening, and after the connected-components area filter. Regenerated from the new ablation runs.

Fig. 9 Mask cleanup on the first canonical input clip at frame 200. The normalized response field (1) is thresholded into a noisy binary mask (2). Closing welds neighbouring wet patches and fills small gaps (3). Opening removes specks the closing kernel could not fill (4). Connected-components labelling removes the few residual islands below the four-hundred-pixel area floor, leaving the final mask on which the temporal lock operates (5).

The kernel parities are forced odd at runtime so that anchor handling is symmetric. The structuring-element shape is elliptical by default with optional rectangular and cross-shaped alternatives, exposed because the front shape changes with infusion geometry.

Morphological-kernel comparison placeholder. Three panels of the cleaned mask under each structuring-element shape with $k_m = 13$. Left: ellipse (used by the integrated configuration), isotropic, appropriate for round fronts. Center: rectangle, asymmetric, appropriate for stripe-shaped wet regions. Right: cross, minimal, preserves corners. The integrated panel is bordered in garnet.

Fig. 10 Morphological structuring-element shape applied to the same threshold output. The integrated configuration uses ellipse.

Table 6 Structuring-element shapes selectable via `--morph-shape`. Used by both the closing and opening passes.

Option	Description
ellipse	Disk-shaped structuring element. Isotropic; appropriate for round fronts.
rect	Rectangular structuring element. Asymmetric; appropriate for stripe-shaped wet regions.
cross	Plus-shaped structuring element. Minimal; preserves corners.

L. Temporal locking

Stage twelve imposes hysteresis along the time axis. Each pixel keeps a small per-pixel counter that increments while the cleaned mask reports the pixel as wet and resets to zero on any frame where the cleaned mask says dry. Once the counter reaches the threshold n_{lock} frames, a sticky locked-pixel map is set true at that location and never resets. The output of the stage is the elementwise OR of the cleaned mask with the sticky locked map, so a pixel that has ever been wet for n_{lock} consecutive frames stays wet for the remainder of the run. Figure Fig. 11 illustrates the bookkeeping with a twelve-frame example for $n_{\mathrm{lock}} = 3$.

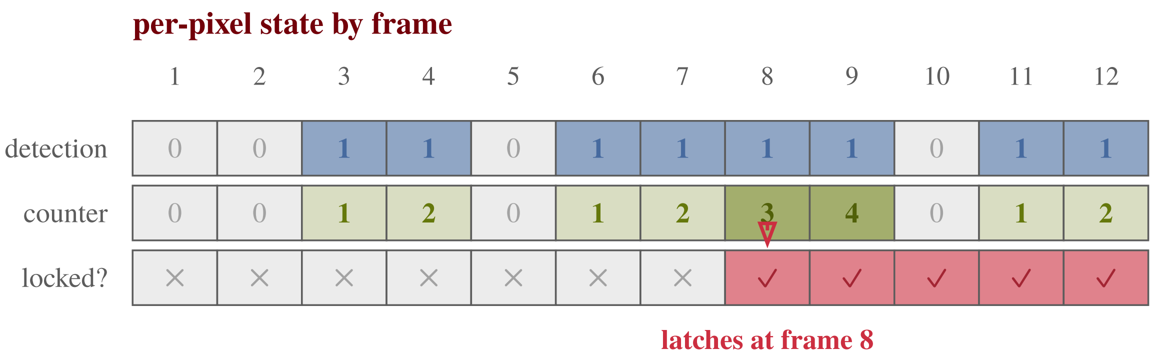

Fig. 11 Lock state for one pixel across twelve frames with $n_{\text{lock}} = 3$. The detection row records what the cleaned mask said this frame. The counter row tracks consecutive wet frames and resets on dry. The locked row latches at the first frame where the counter reaches three (frame eight in this example) and never returns to false. A flicker that survives for fewer than three frames is treated as a transient and not committed.

The integrated configuration uses $n_{\text{lock}} = 3$, which holds a positive detection in the mask for three subsequent frames. The latched-after-three-frames behavior is appropriate for infusions whose true pauses are shorter than the lock window and inappropriate for infusions whose pauses are longer, where the latch produces phantom regions that look like artifacts as discussed in Section 7. Setting the lock window to zero disables the stage altogether. The "no temporal lock" row of Table Table 12 reports the per-sample effect, which depends on whether the live infusion contains pauses on the order of three frames or longer.

M. Heatmap, overlay, and contour export

The last two display stages exist for inspection and label export. The heatmap renderer maps the normalized delta $\tilde{D}_t$ through the Turbo colormap to produce a three-channel pseudocolor image. The overlay renderer extracts the boundary of the locked mask via a five-by-five rectangular morphological gradient and paints those boundary pixels red on a copy of the original BGR frame. Neither renderer is part of the detection logic; they expose, in human-readable form, the field on which the threshold acted and the boundary that the locked mask encloses.

For machine-readable export, the contour-extraction stage emits Common Objects in Context (COCO) and YOLO-format polygons for every connected component of the locked mask. Polygon segmentation is computed with the Suzuki-Abe topological border-following algorithm [22], optionally simplified by the Douglas-Peucker algorithm with tolerance ε, and optionally densified to a maximum edge length so that downstream rasterization preserves curvature. The annotation-sampling utility selects which frames receive labels using one of three modes, namely all-frame, evenly-spaced count, or fixed-stride, with deduplication of consecutive ties so that the integer-truncated linear-spacing exactly reproduces the standard reference behaviour.

Render-output placeholder. Four panels for one frame of the canonical reference video. Panel 1: raw BGR input. Panel 2: heatmap render of the normalized response field $\tilde{D}_t$ through the Turbo colormap. Panel 3: overlay of the locked-mask boundary in red on the original BGR frame. Panel 4: COCO-format polygon export drawn over the input with the polygon vertices marked. Together the four panels show what the pipeline emits for human review and machine-readable export.

Fig. 12 The three render outputs for one frame: heatmap (panel 2), overlay (panel 3), COCO contour export (panel 4). The raw input (panel 1) is shown for reference.

N. Scalar parameters held fixed

Table Table 7 collects the scalar parameter values held fixed across every experiment reported in this paper. The mode menus that each subsection above exposes are documented in their respective tables and are pinned in prose at "the integrated configuration uses…"; the scalars are listed here in one compact table so a reader can recover the exact operating point without scraping ten subsections. The values were chosen during development on a held-out subset disjoint from the labeled evaluation set and were not retuned per video or per metric.

Table 7 Scalar parameter values held fixed in the integrated configuration across every experiment reported in this paper. The only equation that depends on them explicitly is the dynamic-mode lag of Eq. Eq. (1). Mode-menu choices (colorspace, motion-model, peak- on-shift, ref-mode, blur kind, threshold kind, morph-shape) are documented in the per-subsection tables.

Symbol	Value	Where used
`roi-margin`	0.15	Symmetric fractional region-of-interest margin per edge.
`motion-per-frame-threshold`	1.5 px	Per-frame translation magnitude that triggers registration.
`cumulative-motion-threshold`	3 px	Five-frame rolling cumulative drift that triggers registration.
α	0.05	Exponential-moving-average weight for the running reference.
k_p	0	Pre-delta blur kernel size (zero disables the stage).
k_b	9	Post-delta blur kernel size in pixels (Gaussian).
δ_τ	-30	Threshold offset added to Otsu / Triangle / manual / percentile.
k_m	13	Morphological structuring-element size in pixels.
`morph-close-iterations`	1	Closing iterations applied at stage ten.
`morph-open-iterations`	1	Opening iterations applied at stage eleven.
a_{min}	400 px	Minimum connected-component area accepted by the area filter.
n_{lock}	3	Consecutive-wet frames required to latch a pixel in the lock map.
N_{calib}	10	Frames used to fit the dynamic-reference factor κ.
ρ	0.2	Target wet-area fraction for the dynamic-reference lag.
λ	1.0	Multiplicative lag-scale factor in Eq. Eq. (1).
`dynamic-ref-cache-size`	32 frames	Frames cached for the dynamic-reference reader.
`frame-step`	1	Frame stride at decode time.

V. Datasets and Labeling Protocol

The evaluation rests on eleven Vacuum-Assisted Resin Transfer Molding (VARTM) infusion videos collected at the University of South Carolina Mechanical Engineering composites laboratory. Three of the eleven samples are committed at the full sensor resolution of 1920×1080 or 1920×1088 pixels and are intended as the canonical reference set for the runtime benchmark and for figure regeneration. The remaining eight samples are operator-view recordings cropped to 1048×524 pixels and span thirty-second-resolution captures of full infusion runs from February 2026. Sample durations range from twenty-three seconds for the shortest standard-rate clip to roughly nine minutes for the longest cropped run, and frame counts range from one hundred to over fifteen thousand. Two of the high-resolution clips are recorded at a time-lapse rate of 3.33 frames per second, while the remaining nine record at 30 frames per second. The sample inventory is summarized in Table Table 8.

Table 8 Sample inventory. Eleven VARTM infusion clips committed under `data/`. The high-resolution clips serve as the canonical benchmark targets; the cropped operator-view recordings contribute the long-form generalization evidence.

Sample	Frames	FPS	Width	Height	Class
`input_1`	706	29.97	1920	1080	high-res standard
`input_2`	100	3.33	1920	1088	high-res time-lapse
`input_3`	200	3.33	1920	1088	high-res time-lapse
`input_4`	8 100	30.00	1048	524	cropped operator
`input_5`	8 100	30.00	1048	524	cropped operator
`input_6`	8 100	30.00	1048	524	cropped operator
`input_7`	8 100	30.00	1048	524	cropped operator
`input_8`	12 600	30.00	1048	524	cropped operator
`input_9`	15 469	30.00	1048	524	cropped operator
`input_10`	11 426	30.00	1048	524	cropped operator
`input_11`	14 876	30.00	1048	524	cropped operator

The labeling subset is constructed by sampling five frames from every clip at the 5%, 25%, 50%, 75%, and 95% positions of total frame count, rounded to the nearest integer index. The sampling is stratified within each sample so that the fifty-five-frame ground-truth set covers every clip equally rather than concentrating in a single high-frame-count recording. The resulting set spans early-fill, early-mid, mid, mid-late, and late-fill stages within each clip, and across clips it spans the four resolution and frame-rate regimes described above.

A. Labeling protocol

Each anchor frame is labeled by a single human annotator under the protocol committed at `data/LABELING_PROTOCOL.md`. The annotator marks one polygon per frame indicating the resin-wetted region, defined as the area where resin has visibly transitioned the fabric from dry to saturated. Specular reflections from the silicone vacuum bag, vacuum-bag wrinkles and creases that pre-date the front, fabric-weave shadows that pre-date the front, and pooled resin in inlet runners outside the laminate boundary are explicitly excluded. The annotator distinguishes real wetting from transient appearance changes by scrubbing forward a few frames and confirming that the candidate region's brightness has changed monotonically. Real wetting darkens once and stays dark; reflections and wrinkles oscillate. The labeling tool is the browser-based polygon editor at makesense.ai, chosen for its ability to import a flat directory of PNGs and export Common Objects in Context (COCO) JSON without intermediate format conversion. Fifty-five labeled images are exported as a single COCO file at `data/labels.json`. Figure Fig. 13 shows a representative labeled frame.

Labeling protocol illustration placeholder. One representative frame from the fifty-five-frame ground-truth subset, with the operator's polygon overlaid in rose. Replaced once labels arrive.

Fig. 13 Example labeled frame from the labeling pass. The rose polygon indicates the wet region as judged by the human annotator following the criteria in `data/LABELING_PROTOCOL.md`.

B. Scope and known limits of the labeling subset

The labeling subset trades depth for breadth. Five frames per sample across eleven distinct molds is the lowest count at which the per-sample agreement breakdown can surface a sample-specific failure mode. Inter-annotator agreement is not reported because a single annotator labeled the entire subset. The labeling protocol was committed to the repository before the labeling pass began, so the criteria for what counts as wet are recoverable and could be applied by a second annotator in follow-on work to produce a kappa-style agreement number without ambiguity.

VI. Quantitative Agreement Results

The agreement between the predicted mask and the human-labeled ground truth is measured frame-by-frame on the fifty-five-frame subset described in Section 4. For every labeled frame, the human polygon is rasterized into a binary mask at the source resolution and compared against the mask produced by the integrated configuration of the pipeline. Five complementary metrics are reported. Mask Intersection-over-Union (IoU) is the headline accuracy metric. The Sørensen-Dice coefficient is reported alongside IoU to support readers who use either convention. Boundary F_1 is computed as the mean of three pixel-tolerance values $\tau \in \{1, 3, 5\}$ pixels, where the per-tolerance F_1 is the harmonic mean of precision (prediction-boundary pixels within τ pixels of any ground-truth-boundary pixel) and recall (ground-truth-boundary pixels within τ pixels of any prediction-boundary pixel). Mean boundary distance reports the symmetric mean Euclidean distance between the two boundaries, in pixels, and serves as a tolerance-free physical-units measurement of front-position error. Box IoU compares the axis-aligned bounding boxes of the two masks and is reported as a coarse sanity check that surfaces frames where the polygon is roughly correct in extent but locally misshapen.

Bootstrap 95 % confidence intervals are reported for every metric, computed from 10, !000 resamples of the per-frame metric values with replacement. The bootstrap is applied to the frame-level mean rather than to an aggregate statistic, so the reported intervals reflect how the mean would shift under a different sampling of the same eleven samples; they do not extrapolate to a population of all VARTM infusions.

A. Integrated configuration vs. naive baseline

The integrated configuration is the configuration described in Section 3 and held fixed across every result reported in this section, with the mode menus pinned in their respective per-subsection tables and the scalar values listed in Table Table 7. The naive baseline runs the same pipeline with every named component disabled, namely no peak-brightness reference, no darken-only clip, no temporal lock, no morphological cleanup, and no dynamic-lag reference selection. The naive baseline retains only the colorspace projection, the ROI restriction, and the Otsu threshold against the first-frame reference, which is the minimum that any prior camera-based VARTM system reports doing. The contrast between the two rows is the empirical answer to the question "does the joint configuration matter for IoU on this benchmark."

Table 9 Integrated configuration vs. naive baseline on the fifty-five-frame labeling subset. The naive baseline runs the same pipeline with every named component disabled, retaining only the colorspace projection, ROI restriction, and Otsu threshold against the first-frame reference, which is the minimum any prior camera-based VARTM system reports doing. Each cell carries a bootstrap 95 % confidence interval over $10, !000$ resamples of the per-frame mean; intervals omitted from the table for readability and reported in the per-metric breakdown of Table Table 10. B-F_1 is mean boundary F_1 across $\tau \in \{1, 3, 5\}$ pixels.

Configuration	IoU	Dice	B-F_1	Boundary px	Box IoU
Integrated	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*
Naive baseline	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*

Table 10 Integrated-configuration agreement on the fifty-five-frame labeling subset, all five metrics with bootstrap 95 % confidence intervals over $10, !000$ resamples of the per-frame mean.

Metric	Mean	95 % CI
Mask IoU	*TBD*	[*TBD*, *TBD*]
Sørensen-Dice	*TBD*	[*TBD*, *TBD*]
Boundary F_1	*TBD*	[*TBD*, *TBD*]
Mean boundary distance (px)	*TBD*	[*TBD*, *TBD*]
Box IoU	*TBD*	[*TBD*, *TBD*]

B. Per-sample breakdown

The eleven samples in Section 4 differ substantially in resolution, frame rate, illumination, and operator framing. A per-sample breakdown is the strongest available evidence that the agreement reported above is consistent across substantively different molds rather than driven by a single fortunate recording. Table Table 11 reports mask IoU and boundary F_1 for each sample alongside the count of labeled frames contributing to the mean.

Table 11 Per-sample agreement for mask IoU and boundary F_1. Each row aggregates the five anchor frames sampled at the $\frac{\frac{\frac{\frac{5}{25}}{50}}{75}}{95}$% fill positions described in Section 4.

Sample	n	Mask IoU	Boundary F_1
`input_1`	5	*TBD*	*TBD*
`input_2`	5	*TBD*	*TBD*
`input_3`	5	*TBD*	*TBD*
`input_4`	5	*TBD*	*TBD*
`input_5`	5	*TBD*	*TBD*
`input_6`	5	*TBD*	*TBD*
`input_7`	5	*TBD*	*TBD*
`input_8`	5	*TBD*	*TBD*
`input_9`	5	*TBD*	*TBD*
`input_10`	5	*TBD*	*TBD*
`input_11`	5	*TBD*	*TBD*

C. Component-removal ablation

Each row of Table Table 12 holds the integrated configuration described in Section 3 fixed, removes one named component, and reports the resulting mean IoU on the fifty-five-frame subset with a bootstrap 95 % confidence interval. The final column reports the absolute change in mean IoU relative to the integrated configuration in the first row, signed so that a negative number is a drop and a positive number is a rise. Per-sample ΔIoU values for the same rows are reported in the supplementary breakdown referenced from Table Table 11. The reader should not expect the per-sample values to share the sign of the eleven-sample mean. A primitive whose assumption matches the dynamics of one sample can be neutral or counterproductive on a sample with different dynamics, and the per-sample breakdown is the empirical content of the ablation rather than the eleven-sample mean alone.

Table 12 Component-removal ablation on the fifty-five-frame labeling subset. Each row holds the integrated configuration of Section 3 (Tables Table 1 through Table 7) fixed and disables one named component. ΔIoU is the signed change in mean IoU relative to the integrated configuration; negative values are drops, positive values are rises. Rows are listed in the order in which each corresponding primitive is introduced in Section 3, not sorted by ΔIoU, so that the order is independent of the data. Confidence intervals are bootstrap quantiles over 10, !000 resamples of the per-frame mean.

Configuration	IoU	95 % CI	ΔIoU
Integrated	*TBD*	[*TBD*, *TBD*]	–
No peak-brightness reference	*TBD*	[*TBD*, *TBD*]	*TBD*
No temporal lock	*TBD*	[*TBD*, *TBD*]	*TBD*
No darken-only clip	*TBD*	[*TBD*, *TBD*]	*TBD*
No dynamic-lag reference	*TBD*	[*TBD*, *TBD*]	*TBD*
No region-of-interest restriction	*TBD*	[*TBD*, *TBD*]	*TBD*
No morphological cleanup	*TBD*	[*TBD*, *TBD*]	*TBD*
No pre-delta blur	*TBD*	[*TBD*, *TBD*]	*TBD*
No camera-shift registration	*TBD*	[*TBD*, *TBD*]	*TBD*
Grayscale colorspace	*TBD*	[*TBD*, *TBD*]	*TBD*
HSV colorspace	*TBD*	[*TBD*, *TBD*]	*TBD*

Component-removal effect-size figure placeholder. Per component in Table Table 12, the eleven-sample mean Δ IoU bar next to a strip plot of the eleven per-sample ΔIoU values.

Fig. 14 Effect-size companion to Table Table 12. For each row in the table, the left bar is the eleven-sample mean ΔIoU and the right strip is the eleven per-sample ΔIoU values. Strips that cross zero identify primitives that are neutral or counterproductive on at least one sample, which is information the eleven-sample mean alone hides.

D. Qualitative montage

Frame montage placeholder. Three columns per labeled frame (raw input, predicted overlay, human polygon) drawn from four representative samples spanning the resolution and frame-rate regimes documented in Section 4. Replaced once the predicted overlays are regenerated from the integrated configuration.

Fig. 15 Qualitative montage of labeled frames. Each row shows one frame from a different sample (high-resolution standard-rate, high- resolution time-lapse, and two cropped operator-view samples), with raw input, predicted overlay, and human polygon side by side. The figure exists to let the reader judge the IoU number against an actual frame rather than against summary statistics.

VII. Runtime and Systems

The integrated pipeline runs at video rate on a single CPU thread, and the same algorithm has been ported to a CUDA implementation that processes the same eleven samples at substantially higher throughput. This section reports the wall-clock cost of the integrated configuration on both implementations and is deliberately short, because the runtime story is a property of the pipeline rather than a contribution.

A. Hardware

All runtimes are reported on a single host. The CPU is an Apple M-series with N_{cores} performance cores at f_{cpu} GHz, N_{ram} GB unified memory, running macOS V_{os}. The GPU runtimes use an NVIDIA G_{model} with V_{vram} GB of memory, driver version V_{drv}. Decode and encode use the system FFmpeg with the libx264 and libopenh264 codecs, both invoked through the OpenCV `videoio` interface. No frame is decoded twice and no result is cached between runs.

B. Throughput

Table Fig. 16 reports per-sample wall-clock for the integrated configuration on the CPU implementation, on the CUDA implementation, and on a Python reference implementation that mirrors the algorithm stage-for-stage and is included as a runtime baseline rather than as a science contribution. Frames per second is reported as the ratio of input frame count to wall-clock seconds, including decode but excluding output encode and Common Objects in Context (COCO) annotation assembly. The CUDA implementation reaches K frames per second aggregated across the eleven samples, an S-fold speedup over the CPU implementation at the same parameter values.

Three-implementation runtime placeholder. Aggregate per-sample wall-clock and frames per second for the Python reference, the CPU implementation, and the CUDA implementation across the eleven VARTM samples. Replaced once the CUDA close-out completes.

Fig. 16 Per-sample wall-clock for the integrated configuration on three implementations of the same algorithm. Wall-clock includes video decode but excludes output encode and annotation export. Speedups are reported relative to the Python reference at matched parameter values.

C. Validation

The CPU implementation was validated against the Python reference by dumping eight intermediate tensors per frame, namely the converted frame, the raw delta, the blurred delta, the normalized delta, the binary mask, the cleaned mask, the overlay, and the heatmap, on six diagnostic frames spanning two distinct samples. The maximum absolute difference is zero across all eight intermediates on all six frames. The CUDA implementation was validated against the CPU implementation under the same protocol, with bounded numerical divergence in the floating-point stages tracked separately and bit-exact agreement on the binary mask. The validation confirms that the runtime numbers in Table Fig. 16 correspond to the same algorithm at all three operating points, not to three different algorithms that happen to agree on summary statistics.

VIII. Discussion

The integrated pipeline is a working method on the eleven samples evaluated in this paper, not a universal segmenter for arbitrary VARTM video. The failure modes described below are concrete, mechanism-level, and traceable to the specific component whose assumption the input violates.

A. Failure Modes

The temporal-locking window is the primary diagnostic when results appear noisy on long pauses. The integrated configuration sets the window to three frames, which holds a positive detection in the mask for three subsequent frames so transient single-frame dropouts do not flicker the boundary. On infusions whose true wet-front pauses exceed three frames, transient detections from earlier in the run persist past the moment the front recedes and the mask grows phantom regions that look like artifacts. The mechanism is not a defect. It is the locking semantics applied to a sequence whose pauses violate the assumption baked into a fixed three-frame window. The operational consequence is that the window is a per-mold parameter, not a constant suitable for every infusion, and the failure is visible only on long-pause sequences.

The dynamic-reference calibration window is the second sensitive surface. The integrated configuration uses ten calibration frames to fit a square-root-of-area growth model that subsequently lags the reference frame behind the live frame. The assumption baked into that fit is that early fill is representative of the growth law. If the first ten processed frames contain race-tracking, an air pocket, or some other anomaly, the fit picks up a poor reference factor and every downstream frame inherits that error. The remedy is to choose a calibration window that excludes the anomaly or to fall back to one of the static reference modes, both of which are configuration choices rather than numerical defects.

The third failure mode is the inherent ceiling of any tuned classical-CV pipeline. The integrated configuration does not generalize across substantially different fabric types, lighting setups, or vacuum-bag textures without re-tuning. The mode menus and scalar values held fixed in Section 3 (Tables Table 1, Table 5, Table 7) assume a transparent vacuum bag, top-down LED lighting, and a dark mold background, because those are the conditions of the eleven samples in the labeling subset. Off-distribution conditions, such as a heavily textured bag, side lighting, or a bright mold surface, can break the Otsu threshold or push the response distribution outside the range the threshold offset was tuned for. The pipeline exposes a percentile-threshold mode, a manual contrast threshold, and an RGB colorspace switch precisely because no single configuration covers every regime, and a per-mold reconfiguration is the appropriate response. The component-removal ablation in Section 5 reports on conditions where the integrated configuration as evaluated holds.

B. Tradeoffs

The integrated configuration is built on a deliberate set of tradeoffs that a reader should weigh alongside the benchmark numbers.

We trade an end-to-end accuracy ceiling for component-level explainability. A learned segmenter trained on a sufficiently large in-domain corpus would plausibly exceed the boundary F_1 reported in this paper, but no such corpus exists in the public VARTM literature and a research group that wants to characterize a single infusion run next week cannot collect one in time. The component-removal ablation in Section 5 reports the IoU consequence of disabling each named primitive on the eleven-sample subset, which is the property the integrated configuration is designed to expose. A monolithic learned segmenter would conflate those deltas, so a reader interested only in the headline number could replace this pipeline with a sufficiently large convolutional model and lose the per-component evidence. The empirical contribution of this paper is the per-component evidence, not the headline.

We trade automated hyperparameter selection for component-level legibility. The pipeline has no learned end-to-end loss and no automated tuner sweeping fifty parameters in the background. Every parameter that controls the result is named in Section 3 and either documented in a per-subsection mode-menu table or pinned in Table Table 7. A reader who wants to understand why a result looks the way it does on a particular infusion can answer that question by reading the configuration values rather than by introspecting weights. The cost is that an unfamiliar reader who picks an inappropriate value for a per-mold parameter can ruin a run, which is the same risk the temporal-locking failure mode above describes.

The pipeline runs on a CPU on the host described in Section 6 and on a GPU when one is present. Neither implementation is part of the empirical claim; both are reported so that a reader can locate the throughput available on a typical research bench and scale that estimate to longer runs without re-deriving it.

C. Scope and Limits

Several setups fall outside the regime evaluated in this paper. Opaque vacuum bags break the entire pipeline, because the front is no longer visually observable from above and there is nothing for any image-domain method to detect. Multi-resin systems where dye contrast is intentionally absent fall in the same category, since the response image is ultimately a contrast measurement. Very fast infusions, where the front passes through a region of interest in well under a second, push the temporal-locking and morphology kernels past the regime they were tuned for, and the resulting masks lag the true front. Camera motion during infusion is handled by the optional registration stage, which warps the live frame back into the reference coordinate system whenever per-frame or rolling-window drift exceeds the configured threshold. The registration absorbs the bumped-tripod and thermal-creep regimes that otherwise break the comparison between live and reference frame. Setups where the camera moves continuously and at high amplitude, such as a handheld phone walked around the rig, exceed the small-shift assumption the affine warp is fit under and remain out of scope.

These are not weaknesses to be patched. They are the boundary of the application domain. A practitioner facing any of them requires a different tool, and stating that boundary explicitly is more useful than overstating the method's operating envelope.

D. Implications for the Field

The integrated pipeline is intended to be useful to two audiences. A practitioner running a VARTM rig obtains a flow-front segmentation method whose effect on each captured infusion is interpretable component by component, so a per-mold problem (specular bag, off-distribution lighting, fast-fill geometry) can be traced to a specific stage rather than buried in an end-to-end loss. A researcher building permeability-inversion or process-monitoring studies that depend on per-pixel front geometry obtains a method whose individual components are characterized against a labeled benchmark, so substituting any one of them for an alternative method can be evaluated against a fixed comparison point rather than against an opaque baseline.

IX. Conclusion

The integrated pipeline reaches mask IoU $X.XXX$ and boundary F_1 $Y.YYY$ across a 55-frame hand-labeled subset spanning eleven distinct VARTM infusion runs. A per-sample component-removal ablation on the same subset reports the marginal IoU effect of disabling each named primitive in turn, namely the peak-brightness reference, the darken-only delta, the region-of-interest restriction, the dynamic-lag reference selection, the morphological cleanup, and the temporal lock. The marginal effects vary by sample; the per-sample breakdown is the load-bearing piece of the ablation, and the eleven-sample mean is reported alongside it rather than as a substitute for it. The pipeline runs at 30 frames per second on a single CPU. The empirical contribution of this paper is the joint configuration together with the per-sample evidence that names which components are doing the work on which infusion regimes.

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